Vehicle Dynamics and Controls

FINAL PROJECT PRESENTATION

MODEL-FREE SPEED CONTROLLER USING ADE

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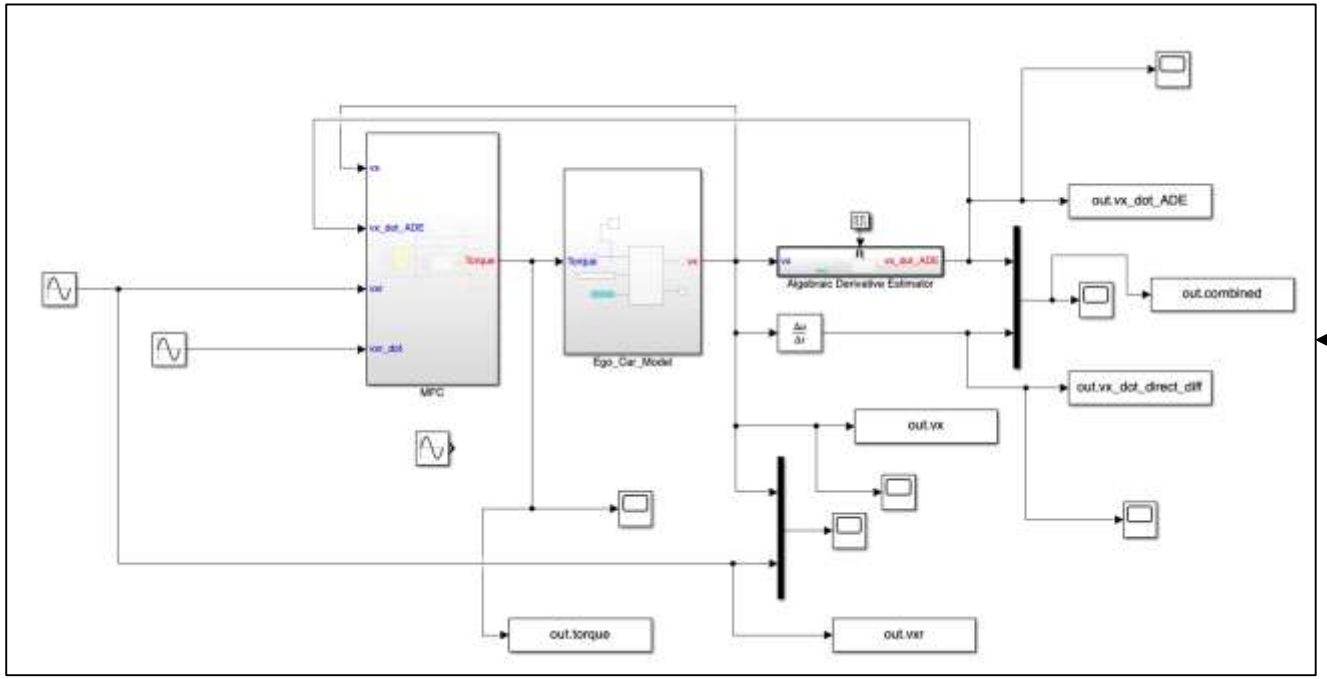
What is MFC?

- Traditional model-based control approaches include PID, adaptive control, and full-state feedback, which generally rely on vehicle parameters that are either difficult to measure accurately or vary with operating conditions.
- **MFC** replaces the unknown vehicle dynamics with an **ultra-local model** - $\dot{v}(t) = F(t) + \alpha u(t)$
- Where $u(t)$ is the input torque, $v(t)$is the longitudinal speed, $F(t)$ combines all unknown dynamics, and α is a chosen scalar representing input effectiveness.
- F(t) is estimated in real time using **Algebraic Derivative Estimation (ADE),** enabling high-performance control without parameter identification.

What is ADE?

- A real-time method that estimates acceleration from noisy speed measurements using algebraic filtering instead of numerical differentiation.
- It provides a smooth, noise-robust acceleration estimate required for stable Model-Free Control.
- ADE is calculated by applying algebraic weights to a sliding window of past velocity samples to compute $\hat{\dot{v}}(t)$.
- $\hat{\dot{v}}(t) = \sum_{k=0}^{N} P_k v(t - kT_s)$



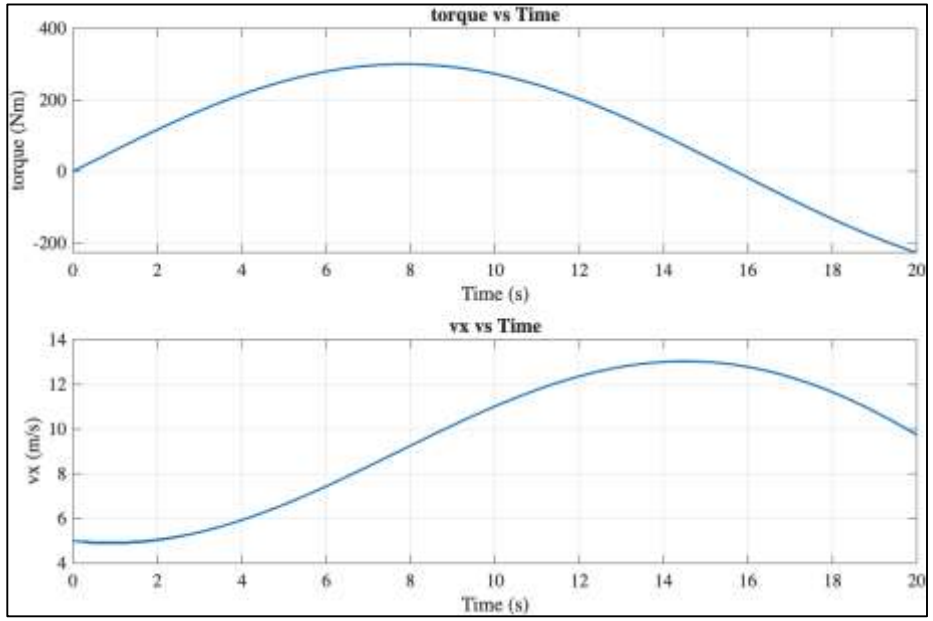
Simulink Model

```
function TORQUE = mfc_torque(v, v_dot_ADE, alpha, vr, vr_dot, T_last)
% Model-Free Speed Controller (unconstrained)
% Inputs:
%   v          - measured speed
%   v_dot_ADE  - ADE estimate of dv/dt
%   alpha      - ultra-local model gain (alpha_v)
%   vr         - reference speed
%   vr_dot     - derivative of reference speed

    % tracking error
    e_v = v - vr;

    % ultra-local model: f_hat = v_dot_ADE
    f_hat = v_dot_ADE - alpha*T_last;

    % tuning gain (choose in script or here)
    eta_T = -5;    % you can tune this value

    % unconstrained torque command
    TORQUE = (-f_hat + vr_dot + eta_T * e_v) / alpha;
end
```

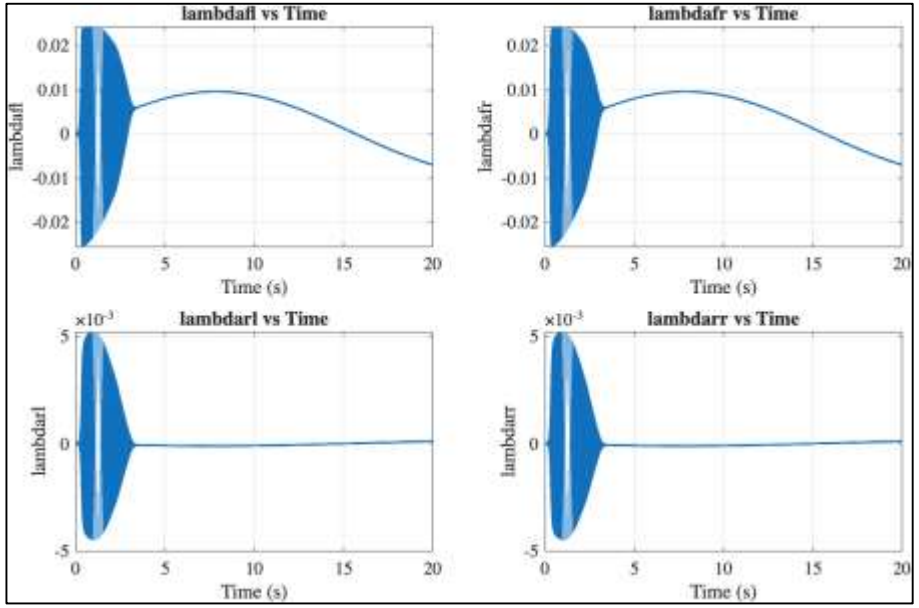
MFC Block

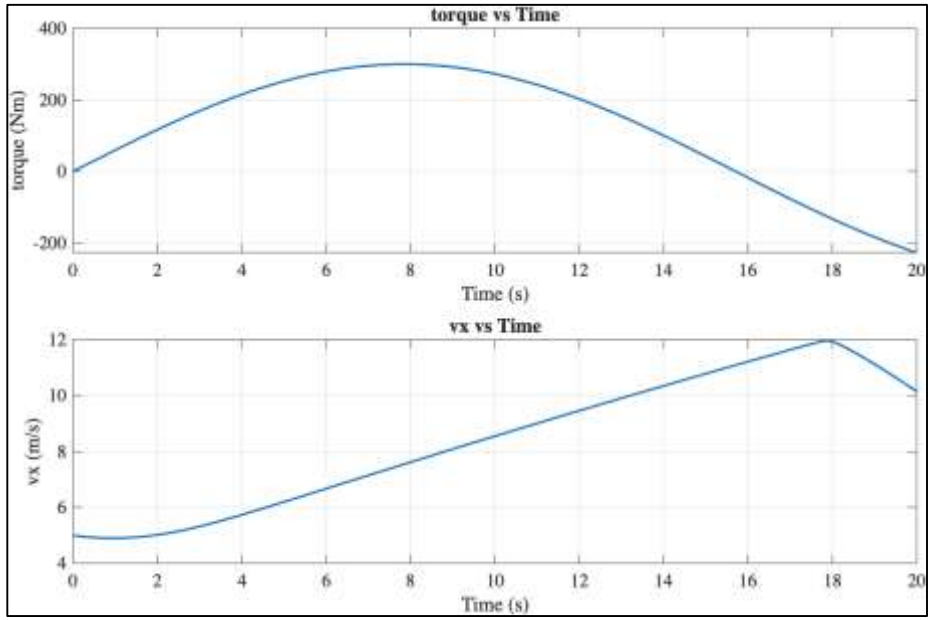
ADE Block

```
function Dot_U = buffer(U, Ts, u_init)
    N =4; % Sliding window length
    persistent y;
    if isempty(y)
        y=u_init*ones(N+1,1); % We initialized a slidi
    end

    y = [U;y(1:end-1)];
    T = Ts*N;
    P = zeros(1,N+1);
    P_alpha = P;
    for k = 1:N+1
        tao_k = (k-1)*Ts;
        P(k) = (6/T^3)*(T - 2*tao_k); % to be filled
        if((k==1)||(k==N+1))
            P_alpha(k) = P(k)*Ts/2;
        else
            P_alpha(k) = P(k)*Ts;
        end
    end

    Dot_U =  P_alpha * y; % to be filled

end
```

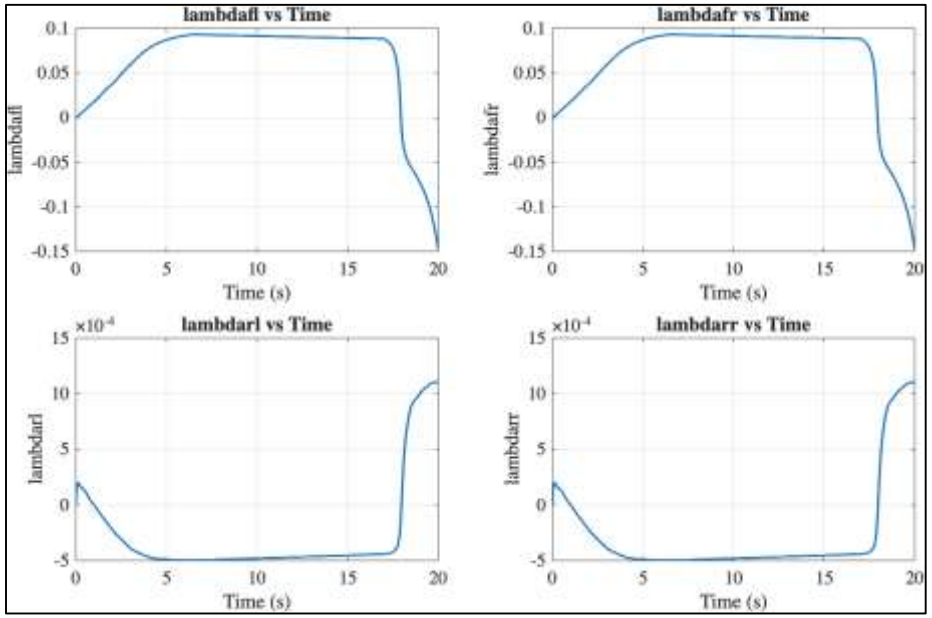


TRFC = 1.0

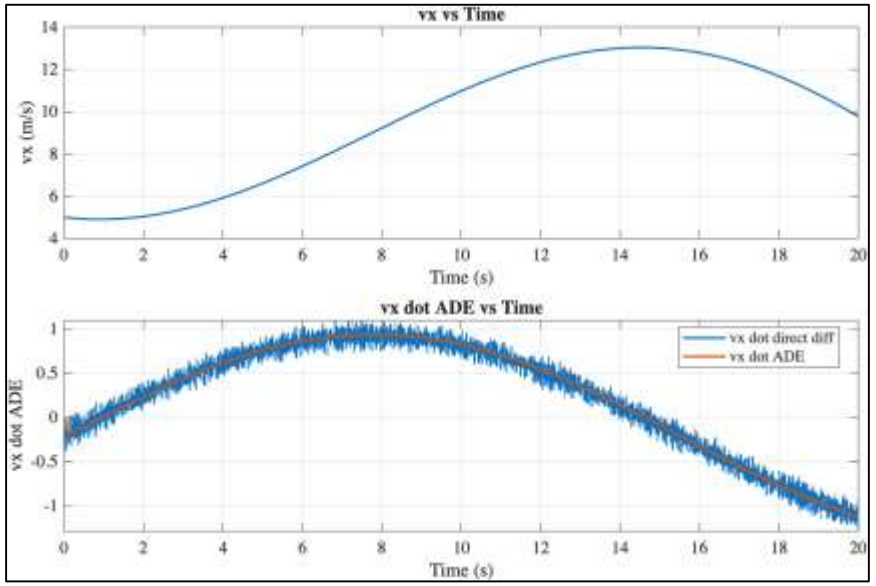


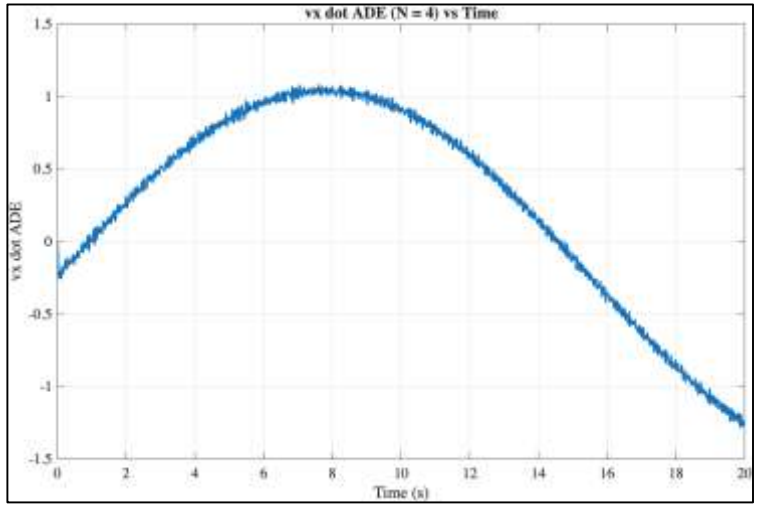


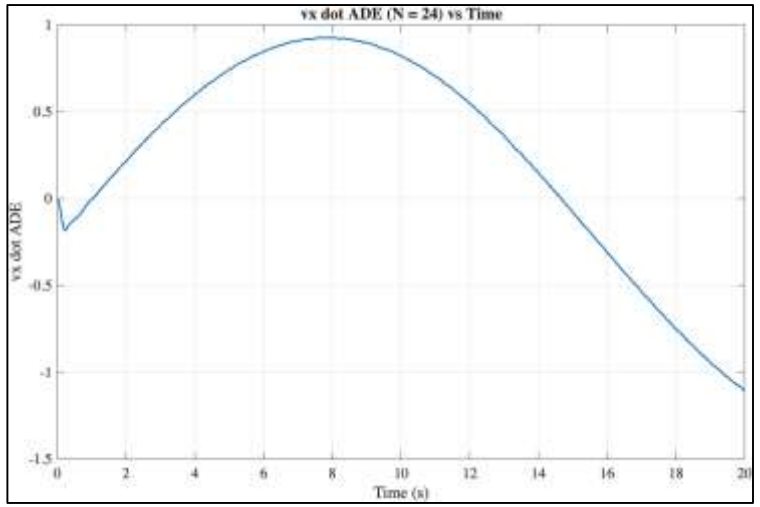
TRFC = 0.1

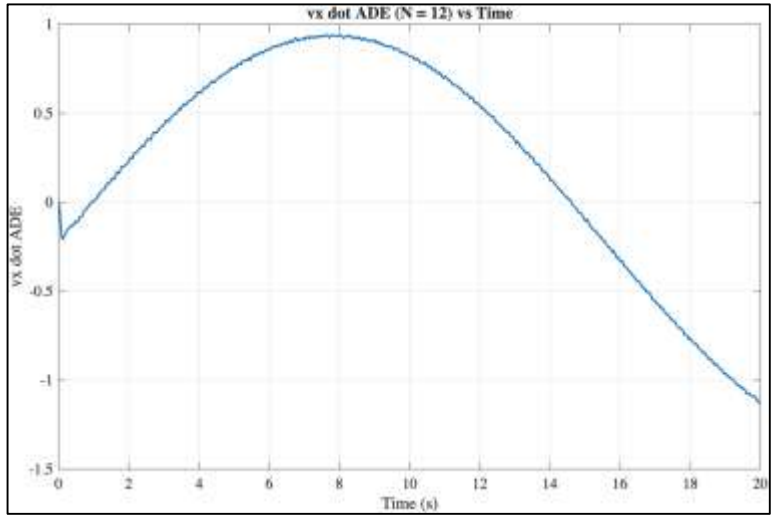


Q2 - Algebraic Derivative Estimation (ADE) performance









Q3: Computing Control Gain α

- Start with wheel rotational dynamics: $J_w \dot{\omega} = T_{rq}$.
- Convert wheel angular acceleration to vehicle acceleration: $\dot{v}_x = R_w \dot{\omega}$.
- Substitute to get physical torque–acceleration relation:

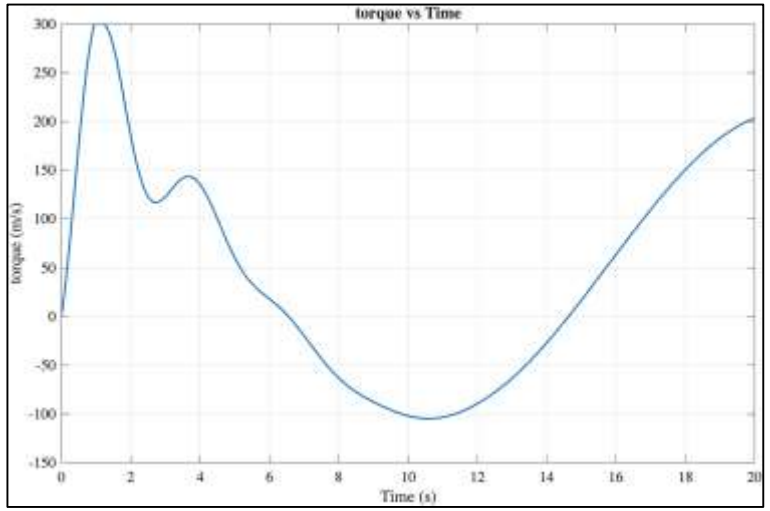
$$\dot{v}_x = \frac{R_w}{J_w} T_{rq}$$

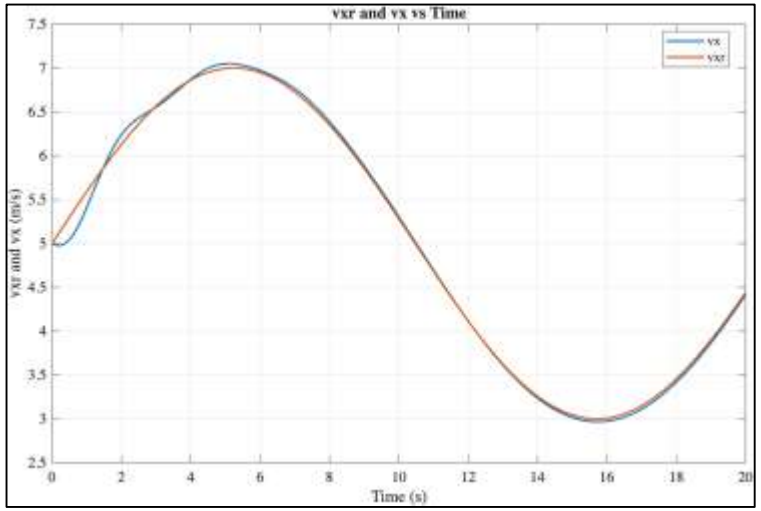
- Therefore, choose control gain:

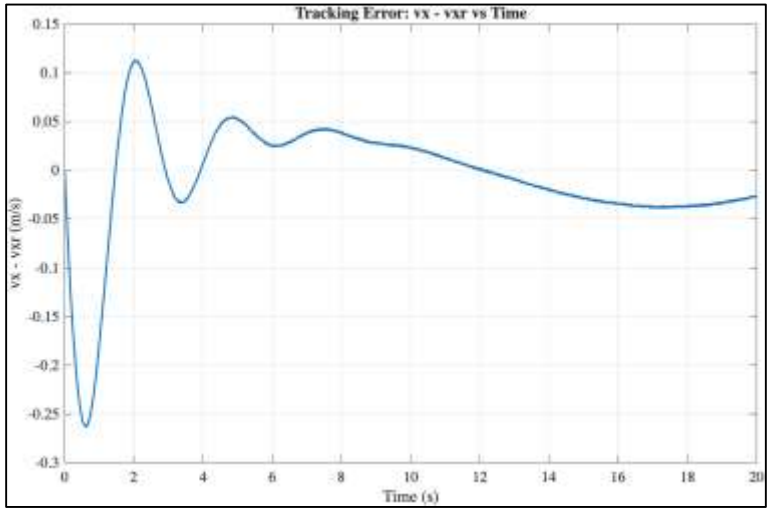
$$\alpha = \frac{R_w}{J_w}$$

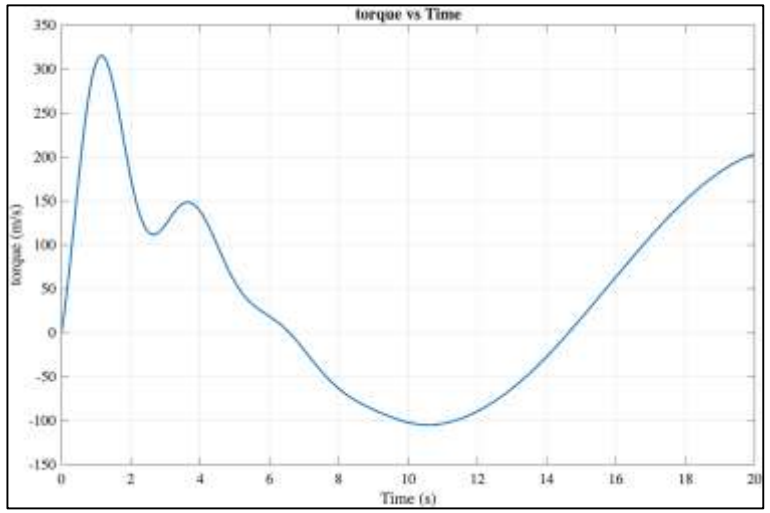
- Using $R_w = 0.33\ m$, $J_w = 0.9\ kg\backslash m^2$,compute: α≈0.37
- This ensures realistic acceleration scaling and stable MFC behavior.

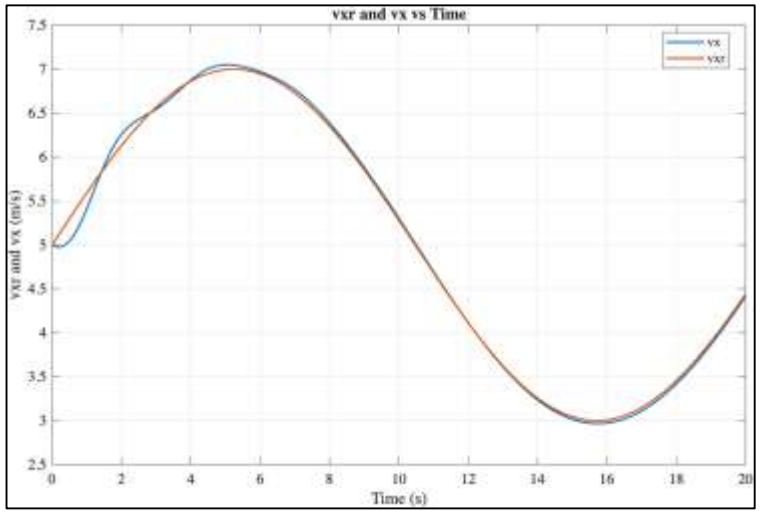
Q4: Tracking Performance with and without torque constraint

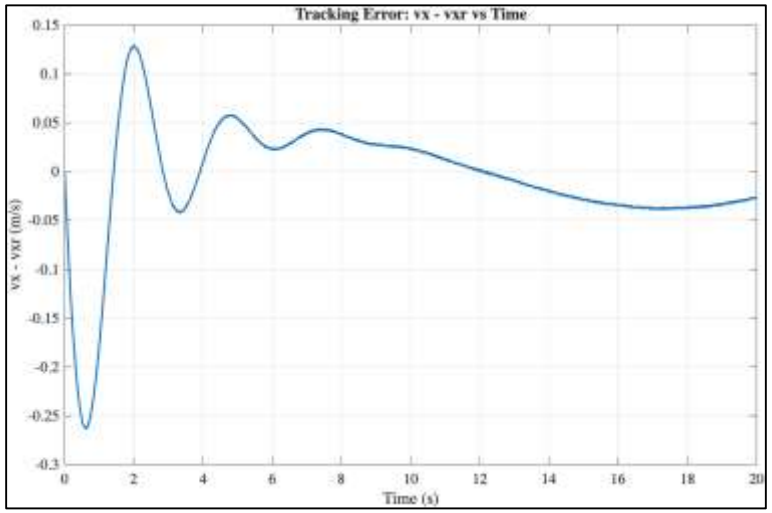












THANK YOU